

Djordje Miladinovic

Curriculum Vitæ

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Summary

- **Profile:** Senior AI/ML engineer | PhD in computer science (AI/ML) | Swiss resident | Serbian citizen. 10+ years in AI/ML research | 15+ years in software development | 5+ years in project leading | 5+ years in life sciences.
- **Work:** Designing and leading the end-to-end development of scalable AI/ML and software products, from concept to deployment.
- **Mission:** My goal is to utilize technology, innovative design, and data-driven insights to improve individual and societal well-being, fuel business growth, and deliver best-in-class solutions with measurable impact.

Experience



Senior CV/ML Engineer, *Reality Labs*

Sep'25-Present

- Developing cutting-edge Spatial AI visual-language reasoning models that power Meta's wearable devices, enabling immersive, intelligent experiences for millions of users worldwide.



Senior AI/ML Engineer, *AI department of GSK plc*

Oct'21-Sep'25

- Led the development of a foundational model integrating multimodal omics data to facilitate preclinical stage of drug discovery – *adopted by three R&D units at GSK, enhancing decision-making in target identification.*
Tech: MLP | Transformer | Representation learning | Large scale data pipelines | Active learning | Counterfactuals;
- Supported clinical efforts to optimize the use of existing GSK drugs.
Tech: GPT-style pretraining | Survival analysis | Random forests | Data from electronic health records (EHRs);



Machine Learning Researcher, *ETH Zürich*

Feb'17-Jun'21

- See the full list of my publications at [Google Scholar](#). Notable projects include:
- **SDN neural networks:** Co-developed a neural architecture for realistic image synthesis – *state-of-the-art performance.*
Tech: Generative image modeling | VAE | LSTM | CNN | Autoregressive models | Representation learning;
- **Sleep-learning web platform:** Led a team of researchers to develop a web-based large-scale machine learning platform for sleep pattern recognition from EEG/EMG signals – *now commercialized with over 100K jobs processed.*
Tech: CNN | Fourier transform | Image classification | Web service;



Machine Learning Researcher (Visitor), *Max Planck Institute for Intelligent Systems*

Sep'18-Dec'18

- Co-developed a [framework for evaluating disentangled representations](#).
Tech: Representation learning | Causal analysis | Robust machine learning;



Machine Learning Researcher & Engineer (Intern), *Logitech Europe S.A.*

Sep'16-Feb'17

- Developed an AI/ML product to predict potential premium users and reduce churn rate.
Tech: Regression | Sentiment analysis | Random forests | SQL;



Machine Learning Researcher (Visitor), *Disney Research Studios, Zurich*

Jan'16-Sep'16

- Developed a software platform for detecting mechanical failures in Disney's humanoid robots based on IMU sensor readings. The platform uses novel machine learning algorithm to asses robot degradation in a human-like way.
Tech: Kalman filter | Motion capture | IMUs | Dynamic time warping | Time series classification | 3D rendering;

Education



PhD in Computer Science – Machine Learning, [ETH Zürich](#)

Sep'17-Jun'21

- *Thesis:* "On training Deep Generative Models with Latent Variables" – [Prof. Joachim M. Buhmann](#).
- *Research areas:* Generative image and text modeling | Variational autoencoders | Representation learning | Deep learning | Large language models | Machine learning for biology;



MS in Computer Science – Machine Learning, [ETH Zürich](#)

Sep'13-Jun'16

- *Thesis:* "Perceptual Analysis Framework for Discovering Anomalies in Humanoid Motions" – [Prof. Otmar Hilliges](#).
- *Focus:* Machine learning | Software engineering | Operating systems | Distributed computing;



BS in Computer Science – Software Engineering, [ETF Belgrade](#)

Sep'09-Sep'13

- *Thesis:* "Java-based Interactive Software Simulator of a CISC Processor" – [Prof. Zaharije Radivojević](#).
- *Focus:* Software engineering and design | Algorithms and data structures | Operating systems | Hardware design;

Technical Skills

Programming	<i>Advanced:</i> Python (PyTorch); <i>Good but rusty:</i> C/C++ Java Javascript Bash SQL;
Infrastructure	<i>Data formats:</i> Parquet H5 H5AD CSV YAML JSON Pickle;
(Varying levels)	<i>Databases:</i> PostgreSQL MongoDB SQLite Milvus; <i>Big data:</i> Spark Hive Hadoop; <i>AI/ML frameworks:</i> Pytorch Lightning Ray Weights & Biases MLflow HuggingFace Kubeflow; <i>High Performance Computing (HPC):</i> SLURM LSF Condor; <i>Middleware:</i> Celery Redis Kafka; <i>Cloud:</i> GCP/GCS AWS/S3 Microsoft Azure; <i>Deployment:</i> Docker Kubernetes; <i>CI/CD:</i> GitHub Actions CircleCI; <i>Web:</i> Node.js Django HTML/CSS;
Python libraries	<i>Data Science:</i> Pandas SciPy Seaborn NumPy Scanpy Scrapy OpenCV; <i>AI/ML:</i> PyTorch Scikit-learn SpaCy Fairseq LangChain Tensorflow;
Design tools	Figma Inkscape Blender PowerPoint Adobe Illustrator;

Languages

Serbian (Native) | English (Fluent) | German (Intermediate) | Spanish (Beginner);

Hobbies

Waterpolo | Swimming | Basketball | Tennis | Chess | Cinematography | Reading;